Osbert Bastani

Employment

2025– **Associate Professor in Computer and Information Sciences**, *University of Pennsylvania*, Philadelphia, PA.

2022–2025 **Assistant Professor in Computer and Information Sciences**, *University of Pennsylvania*, Philadelphia, PA.

2018–2022 **Research Assistant Professor in Computer and Information Sciences**, *University of Pennsylvania*, Philadelphia, PA.

2017–2018 **Postdoctoral Fellow in CSAIL**, *Massachusetts Institute of Technology*, Cambridge, MA.

Education

2012–2018 **Ph.D. in Computer Science**, *Stanford University*, Stanford, CA.

2008–2012 **A.B. in Mathematics**, *Harvard University*, Cambridge, MA.

Awards & Honors

2025 **Cited in 2025 Economic Report of the President**.

2024 **IEEE Micro Top Picks from the Computer Architecture Conferences**.

2024 **NSF CAREER Award**.

2024 **ICRA Finalist for Best Paper Award in Robot Vision**.

2024 **ICRA Best Paper Award (Open X-Embodiment Collaboration)**.

2023 **Amazon Research Award**.

2023 **CoRL LEAP Workshop Best Paper Award**.

2023 **ICML TEACH Workshop Best Paper Award**.

2018 **PLDI Distinguished Paper Award**.

2015–2017 **Google Ph.D. Fellowship**.

Publications

Marcus J. Min, One An, Xujie Si, and Osbert Bastani. Ground False: Uncovering Errors in Formal Mathematics Benchmarks. *NeurIPS (Evaluations and Datasets)*, 2026.

Yuming Feng, Frederick Pu, One An, Osbert Bastani, Li Zhang, Jiani Huang, Xujie Si, and Ziyang Li. Theory-Scale Auto-Formalization of Logics for Computer Science. *NeurIPS (Evaluations and Datasets)*, 2026.

Haydn Thomas Jones, Yimeng Zeng, Alden Rose, Li S. Yifei, Yining Huang, Kaiwen Wu, Jiaming Liang, Maggie Ziyu Huan, Yoseph Barash, Cesar de la Fuente-Nunez, Osbert Bastani, Zachary Ives, Mark Yatskar, and Jacob R. Gardner. Self Driving Datasets: From 20 Million Papers to Nuanced Biomedical Knowledge at Scale. *NeurIPS (Evaluations and Datasets)*, 2026.

Xiaomeng Wang, Hamsa Bastani, Osbert Bastani, and Zhimei Ren. Learning to target with network interference. *NeurIPS*, 2026.

Marcus J. Min, Mike He, Zhaoyu Li, Zixuan Yi, Sharad Malik, Aarti Gupta, Xujie Si, and Osbert

Bastani. Theory-Level Autoformalization: From Isolated Statements to Unified Formal Knowledge Bases. *ICML (Position, Spotlight)*, 2026.

Haosen Ge, Hamsa Bastani, and Osbert Bastani. Position: AI Capabilities May Not Increase Exponentially. *ICML (Position)*, 2026.

Stephen Mell, Botong Zhang, David Mell, Shuo Li, Ramya Ramalingam, Nathan Yu, Stephan Zdancewic, and Osbert Bastani. Quasar: A Programming Language Specialized for LLM Code Actions. *COLM*, 2026.

Ramya Ramalingam, Osbert Bastani, and Aaron Roth. Prior-Agnostic Incentive-Compatible Exploration. *EC*, 2026.

Angel Chung, Patrick Bayoh, Jatu Abdulai, Lawrence Sandi, Francis Smart, Hamsa Bastani, and Osbert Bastani. Improving access to essential medicines in sierra leone via decision-aware machine learning. *Nature*, 2026.

Celeste Barnaby, Danny Ding, Osbert Bastani, and Isil Dillig. Choose, don't label: Multiple-choice query synthesis for program disambiguation. *PLDI*, 2026.

Sagnik Anupam, Alexander Shypula, and Osbert Bastani. Llm program optimization via retrieval augmented search. *ACL (Findings)*, 2026.

Rajeev Alur, Suguman Bansal, Osbert Bastani, and Kishor Jothimurugan. Specification-guided reinforcement learning. *Communications of the ACM*, 69(2):80–87, 2026.

Marcus J Min, Yeqi Gao, Wilson Sy, Zhaoyu Li, Xujie Si, and Osbert Bastani. Divide and abstract: Autoformalization via decomposition and abstraction learning. *ICLR*, 2026.

Michael S Yao, Osbert Bastani, Alma Andersson, Tommaso Biancalani, AÃŊcha Bentaieb, and Claudia Iriondo. Knowledgeable language models as black-box optimizers for personalized medicine. *ICLR*, 2026.

Yimeng Zeng, Natalie Maus, Haydn Thomas Jones, Jeffrey Tao, Fangping Wan, Marcelo Der Torossian Torres, Cesar de la Fuente-Nunez, Ryan Marcus, Osbert Bastani, and Jacob R Gardner. Scaling multi-task bayesian optimization with large language models. *ICLR*, 2026.

William Liang, Sam Wang, Hung-Ju Wang, Osbert Bastani, Yecheng Jason Ma, and Dinesh Jayaraman. Autonomous play with correspondence-driven trajectory warping. *ICLR*, 2026.

Kavi Gupta, Osbert Bastani, and Armando Solar-Lezama. Sparling: End-to-end spatial concept learning via extremely sparse activations. *ICLR*, 2026.

Shuo Li, Rebecca Salowe, Roy Lee, Gui-shuang Ying, Insup Lee, Joan O'Brien, and Osbert Bastani. Development of deep learning model to screen for primary open-angle glaucoma in african ancestry individuals. *npj Digital Medicine*, 2026.

Wenwen Si, Sooyong Jang, Insup Lee, and Osbert Bastani. Conformal constrained policy optimization for cost-effective llm agents. *AAAI*, 2026.

Botong Zhang, Shuo Li, Ignacio Hounie, Osbert Bastani, Dongsheng Ding, and Alejandro Ribeiro. Alignment of large language models with constrained learning. *NeurIPS*, 2025.

Stephen Mell, Konstantinos Kallas, Steve Zdancewic, and Osbert Bastani. Opportunistically parallel lambda calculus. *OOPSLA*, 2025.

Celeste Barnaby, Qiaochu Chen, Ramya Ramalingam, Osbert Bastani, and Isil Dillig. Active learning for neurosymbolic program synthesis. *OOPSLA*, 2025.

Hamsa Bastani, Osbert Bastani, and Wichinpong Park Sinchaisri. Improving human sequential decision making with reinforcement learning. *Management Science*, 72(1):733–755, 2026.

Alexander Shypula, Shuo Li, Botong Zhang, Vishakh Padmakumar, Kayo Yin, and Osbert Bastani. Evaluating the diversity and quality of llm generated content. *COLM*, 2025.

Michael S Yao, Allison Chae, Piya Saraiya, Charles E Kahn Jr, Walter R Witschey, James C Gee, Hersh Sagreiya, and Osbert Bastani. Evaluating acute image ordering for real-world patient cases via language model alignment with radiological guidelines. *Communications Medicine*, 5(1):332, 2025.

Alexander Shypula, Aman Madaan, Yimeng Zeng, Uri Alon, Jacob Gardner, Yiming Yang, Milad Hashemi, Graham Neubig, Parthasarathy Ranganathan, Osbert Bastani, and Amir Yazdanbakhsh. Automated high-level code optimization for warehouse performance. *IEEE Micro Top Picks*, 2025.

Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and Rei Mariman. Generative ai without guardrails can harm learning: Evidence from high school mathematics. *Proceedings of the National Academy of Sciences*, 122(26):e2422633122, 2025.

Michael Yao, James Gee, and Osbert Bastani. Diversity by design: Leveraging distribution matching for offline model-based optimization. *ICML*, 2025.

Haosen Ge, Hamsa Bastani, and Osbert Bastani. Stochastic online conformal prediction with semi-bandit feedback. *ICML*, 2025.

Botong Zhang, Shuo Li, and Osbert Bastani. Conformal structured prediction. *ICLR*, 2025.

Yecheng Jason Ma, Joey Hejna, Chuyuan Fu, Dhruv Shah, Dhruv Liang, Zhuo Xu, Sean Kirmani, Peng Xu, Danny Driess, Ted Xiao, Osbert Bastani, Dinesh Jayaraman, Wenhao Yu, Tingnan Zhang, Dorsa Sadigh, and Fei Xia. Vision language models are in-context value learners. *ICLR*, 2025.

Wentao Guo, Jikai Long, Yimeng Zeng, Zirui Liu, Xinyu Yang, Yide Ran, Jacob R. Gardner, Osbert Bastani, Christopher De Sa, Xiaodong Yu, Beidi Chen, and Zhaozhuo Xu. Zeroth-order fine-tuning of llms with transferable static sparsity. *ICLR*, 2025.

Michael S. Yao, Yimeng Zeng, Hamsa Bastani, Jacob Gardner, James C. Gee, and Osbert Bastani. Generative adversarial model-based optimization via source critic regularization. *NeurIPS*, 2024.

Xinmeng Huang, Shuo Li, Edgar Dobriban, Osbert Bastani, Hamed Hassani, and Dongsheng Ding. One-shot safety alignment for large language models via optimal dualization. *NeurIPS*, 2024.

Xinmeng Huang, Shuo Li, Hamed Hassani, Insup Lee, Osbert Bastani, and Edgar Dobriban. Uncertainty in language models: Assessment through rank-calibration. *EMNLP*, 2024.

William Liang, Sam Wang, Hung-Ju Wang, Jason Ma, Osbert Bastani, and Dinesh Jayaraman. Environment curriculum generation via large language models. *CoRL (Oral)*, 2024.

Kan Xu, Hamsa Bastani, Surbhi Goel, and Osbert Bastani. Stochastic bandits with relu neural networks. *ICML*, 2024.

Jason Ma, William Liang, Hung-Ju Wang, Yuke Zhu, Linxi Fan, Osbert Bastani, and Dinesh Jayaraman. Dreureka: Language model guided sim-to-real transfer. *RSS*, 2024.

DROID Dataset Team. Droid: A large-scale in-the-wild robot manipulation dataset. *RSS*, 2024.

Shuo Li, Sangdon Park, Insup Lee, and Osbert Bastani. Traq: Trustworthy retrieval augmented question answering via conformal prediction. *NAACL*, 2024.

Jason Ma, William Liang, Guanzhi Wang, De-An Huang, Osbert Bastani, Dinesh Jayaraman, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Eureka: Human-level reward design via coding large language models. *ICLR (Spotlight)*, 2024.

Alexander Shypula, Aman Madaan, Yimeng Zeng, Uri Alon, Jacob Gardner, Yiming Yang, Milad Hashemi, Graham Neubig, Parthasarathy Ranganathan, Osbert Bastani, and Amir Yazdanbakhsh. Learning performance-improving code edits. *ICLR (Spotlight)*, 2024.

Wenwen Si, Sangdon Park, Insup Lee, Edgar Dobriban, and Osbert Bastani. Pac prediction sets under label shift. *ICLR*, 2024.

Charles Zhang, Yunshuang Li, Osbert Bastani, Abhishek Gupta, Dinesh Jayaraman, Jason Ma, and Lucas Weihs. Universal visual decomposer: Long-horizon manipulation made easy. *ICRA*, 2024.

Open X-Embodiment Collaboration. Open x-embodiment: Robotic learning datasets and rt-x models. *ICRA*, 2024.

Kavi Gupta, Chenxi Yang, Kayla McCue, Osbert Bastani, Phillip A Sharp, Christopher B Burge, and Armando Solar-Lezama. Improved modeling of rna-binding protein motifs in an interpretable neural model of RNA splicing. *Genome Biology*, 25(1):23, 2024.

Haosen Ge, Hamsa Bastani, and Osbert Bastani. Rethinking fairness for human-ai collaboration. *ITCS*, 2024.

Stephen Mell, Steve Zdancewic, and Osbert Bastani. Optimal program synthesis via abstract interpretation. *POPL*, 2024.

Yahan Yang, Sunghye Cho, Maxine Covello, Azia Knox, Osbert Bastani, James Weimer, Edgar Dobriban, Robert Schultz, Insup Lee, and Julia Parish-morris. Automatically predicting perceived conversation quality in a pediatric sample enriched for autism. *INTERSPEECH*, 2023.

Yanju Chen, Chenglong Wang, Xinyu Wang, Osbert Bastani, and Yu Feng. Fast and reliable program synthesis via user interaction. *ASE*, 2023.

Adam Khakhar, Stephen Mell, and Osbert Bastani. PAC prediction sets for large language models of code. *ICML*, 2023.

Yecheng Jason Ma, Vikash Kumar, Amy Zhang, Osbert Bastani, and Dinesh Jayaraman. LIV: Language-image representations and rewards for robotic control. *ICML*, 2023.

Kishor Jothimurugan, Steve Hsu, Osbert Bastani, and Rajeev Alur. Robust subtask learning for compositional generalization. *ICML*, 2023.

Rajeev Alur, Osbert Bastani, Kishor Jothimurugan, Mateo Perez, Fabio Somenzi, and Ashutosh Trivedi. Policy synthesis and reinforcement learning for discounted LTL. *CAV*, 2023.

Stephen Mell, Favyen Bastani, Steve Zdancewic, and Osbert Bastani. Synthesizing trajectory queries from examples. *CAV*, 2023.

Jason Yecheng Ma, Kausik Sivakumar, Jason Yan, Osbert Bastani, and Dinesh Jayaraman. Learning policy-aware models for model-based reinforcement learning via transition occupancy matching. *L4DC*, 2023.

Wenwen Si, Shuo Li, Sangdon Park, Insup Lee, and Osbert Bastani. Angelic patches for improving third-party object detector performance. *CVPR*, 2023.

Sangdon Park, Osbert Bastani, and Taesoo Kim. Acon2: Adaptive conformal consensus for provable blockchain oracles. *USENIX Security*, 2023.

Wanqiao Xu, Jason Yecheng Ma, Kan Xu, Hamsa Bastani, and Osbert Bastani. Uniformly conservative exploration in reinforcement learning. *AISTATS*, 2023.

Jason Yecheng Ma, Shagun Sodhani, Dinesh Jayaraman, Osbert Bastani, Vikash Kumar, and Amy Zhang. VIP: Towards universal visual reward and representation via value-implicit pre-training. *ICLR (Spotlight)*, 2023.

Jason Yecheng Ma, Jason Yan, Dinesh Jayaraman, and Osbert Bastani. Offline goal-conditioned reinforcement learning via f-advantage regression. *NeurIPS*, 2022.

Osbert Bastani, Varun Gupta, Christopher Jung, Georgy Noarov, Ramya Ramalingam, and Aaron Roth. Practical adversarial multivalid conformal prediction. *NeurIPS (Oral)*, 2022.

Osbert Bastani, Jason Ma, Estelle Shen, and Wanqiao Xu. Regret bounds for risk-sensitive reinforcement learning. *NeurIPS*, 2022.

Halley Young, Maxwell Du, and Osbert Bastani. Neurosymbolic deep generative models for sequence data with relational constraints. *NeurIPS*, 2022.

Sangdon Park, Edgar Dobriban, Insup Lee, and Osbert Bastani. Pac prediction sets for meta-learning. *NeurIPS*, 2022.

Souradeep Dutta, Kaustubh Sridhar, Osbert Bastani, Edgar Dobriban, James Weimer, Insup Lee, and Julia Parish-Morris. Exploring with sticky mittens: Reinforcement learning with expert interventions via option templates. *CoRL*, 2022.

Soham Dan, Osbert Bastani, and Dan Roth. Understanding robust generalization in learning regular languages. *ICML*, 2022.

Sooyong Jang, Sangdon Park, Insup Lee, and Osbert Bastani. Sequential covariate shift detection using classifier two-sample tests. *ICML*, 2022.

Jason Yecheng Ma, Andrew Shen, Dinesh Jayaraman, and Osbert Bastani. Smodice: Versatile offline imitation learning via state occupancy matching. *ICML*, 2022.

Kishor Jothimurugan, Suguman Bansal, Osbert Bastani, and Rajeev Alur. Specification-guided learning of nash equilibria with high social welfare. *CAV*, 2022.

George Tolkachev, Stephen Mell, Steve Zdancewic, and Osbert Bastani. Counterfactual explanations for natural language interfaces. *ACL (Short)*, 2022.

Sangdon Park, Edgar Dobriban, Insup Lee, and Osbert Bastani. PAC prediction sets under covariate shift. *ICLR*, 2022.

Jason Y. Ma, Andrew Shen, Osbert Bastani, and Dinesh Jayaraman. Conservative and adaptive penalty for model-based safe reinforcement learning. *AAAI*, 2022.

Jason Y. Ma, Dinesh Jayaraman, and Osbert Bastani. Conservative offline distributional reinforcement learning. *NeurIPS*, 2021.

Yichen Yang, Jeevana P. Inala, Osbert Bastani, Yewen Pu, Armando Solar-Lezama, and Martin Rinard. Program synthesis guided reinforcement learning for partially observed environments. *NeurIPS (Spotlight)*, 2021.

Kishor Jothimurugan, Suguman Bansal, Osbert Bastani, and Rajeev Alur. Compositional reinforcement learning from logical specifications. *NeurIPS*, 2021.

Alexis Ross, Himabindu Lakkaraju, and Osbert Bastani. Learning models for actionable recourse. *NeurIPS*, 2021.

Soham Dan, Osbert Bastani, and Dan Roth. Few-shot novel concept learning for semantic parsing. *Findings of EMNLP*, 2021.

Jason Y. Ma, Jeevana I. Priya, Dinesh Jayaraman, and Osbert Bastani. Likelihood-based diverse sampling for trajectory forecasting. *ICCV*, 2021.

Favyen Bastani, Songtao He, Ziwen Jiang, Osbert Bastani, and Sam Madden. Skyquery: An aerial drone video sensing platform. *Onward*, 2021.

Radoslav Ivanov, Kishor Jothimurugan, Steve Hsu, Vaidya Shaan, Rajeev Alur, and Osbert Bastani. Compositional learning and verification of neural network controllers. *EMSOFT*, 2021.

Osbert Bastani, Shuo Li, and Anton Xue. Safe reinforcement learning via statistical model predictive shielding. *RSS*, 2021.

Kan Xu, Xuanyi Zhao, Hamsa Bastani, and Osbert Bastani. Group-sparse matrix factorization for transfer learning of word embeddings. *ICML*, 2021.

Jocelyn Chen, Aaron Lamoreaux, Xinyu Wang, Greg Durrett, Osbert Bastani, and Isil Dillig. Web question answering with neurosymbolic program synthesis. *PLDI*, 2021.

Osbert Bastani. Safe reinforcement learning with nonlinear dynamics via model predictive shielding. *ACC*, 2021.

Kishor Jothimurugan, Osbert Bastani, and Rajeev Alur. Abstract value iteration for hierarchical deep reinforcement learning. *AISTATS*, 2021.

Min Wen, Osbert Bastani, and Ufuk Topcu. Algorithms for fairness in sequential decision making. *AISTATS*, 2021.

Sangdon Park, Shuo Li, Insup Lee, and Osbert Bastani. PAC confidence predictions for deep neural network classifiers. *ICLR*, 2021.

Jeevana P. Inala, Yichen Yang, James Paulos, Yewen Pu, Osbert Bastani, Vijay Kumar, Martin Rinard, and Armando Solar-Lezama. Neurosymbolic transformers for multi-agent communication. *NeurIPS*, 2020.

Jiani Huang, Calvin Smith, Osbert Bastani, Rishabh Singh, Aws Albarghouthi, and Mayur Naik. Generating programmatic referring expressions via program synthesis. *ICML*, 2020.

Himabindu Lakkaraju, Nino Arsov, and Osbert Bastani. Robust and stable black box explanations. *ICML*, 2020.

Yanju Chen, Chenglong Wang, Osbert Bastani, Isil Dillig, and Yu Feng. Program synthesis using deduction-guided reinforcement learning. *CAV*, 2020.

Shuo Li and Osbert Bastani. Robust model predictive shielding for safe reinforcement learning with stochastic dynamics. *ICRA*, 2020.

Osbert Bastani. Sample complexity of estimating the policy gradient for nearly deterministic dynamical systems. *AISTATS*, 2020.

Sangdon Park, Osbert Bastani, Jim Weimer, and Insup Lee. Calibrated prediction with covariate shift via unsupervised domain adaptation. *AISTATS*, 2020.

Sangdon Park, Osbert Bastani, Nikolai Matni, and Insup Lee. PAC confidence sets for deep neural networks via calibrated prediction. *ICLR*, 2020.

Jeevana P. Inala, Osbert Bastani, Zenna Tavares, and Armando Solar-Lezama. Synthesizing programmatic policies that inductively generalize. *ICLR*, 2020.

Himabindu Lakkaraju and Osbert Bastani. "How do I fool you?": Manipulating user trust via misleading black box explanations. *AIES*, 2020.

Kishor Jothimurugan, Rajeev Alur, and Osbert Bastani. Composable specifications for reinforcement learning. *NeurIPS*, 2019.

Osbert Bastani, Xin Zhang, and Armando Solar-Lezama. Verifying fairness properties via concentration. *OOPSLA*, 2019.

Jai Chen, Jiayi Wei, Yu Feng, Osbert Bastani, and Isil Dillig. Relational verification using reinforcement learning. *OOPSLA*, 2019.

Zhengkai We, Evan Johnson, Wei Yang, Osbert Bastani, Dawn Song, Jian Peng, and Tao Xie. Reinam: Reinforcement learning for input-grammar inference. *FSE*, 2019.

Arbaaz Khan, Chi Zhang, Shuo Li, Jiayue Wu, Brent Schlotfeldt, Sarah Tang, Alejandro Ribeiro, Osbert Bastani, and Vijay Kumar. Learning safe unlabeled multi-robot planning with motion constraints. *IROS*, 2019.

Halley Young, Osbert Bastani, and Mayur Naik. Learning neurosymbolic generative models via program synthesis. *ICML*, 2019.

Osbert Bastani, Yewen Pu, and Armando Solar-Lezama. Verifiable reinforcement learning via policy extraction. *NeurIPS*, 2018.

Osbert Bastani, Rahul Sharma, Alex Aiken, and Percy Liang. Active learning of points-to specifications. *PLDI*, 2018.

Yu Feng, Ruben Martins, Osbert Bastani, and Isil Dillig. Program synthesis using conflict-driven learning. *PLDI*, 2018.

Osbert Bastani, Carolyn Kim, and Hamsa Bastani. Interpretability via model extraction. *FAT/ML*, 2017.

Osbert Bastani, Rahul Sharma, Alex Aiken, and Percy Liang. Synthesizing program input grammars. *PLDI*, 2017.

Yu Feng, Osbert Bastani, Ruben Martins, Isil Dillig, and Saswat Anand. Automated synthesis of semantic malware signatures using maximum satisfiability. *NDSS*, 2017.

Osbert Bastani, Yani Ioannou, Lenonidas Lampropoulos, Dimitrios Vytiniotis, Aditya Nori, and Antonio Criminisi. Measuring neural net robustness with constraints. *NIPS*, 2016.

Lazaro Clapp, Osbert Bastani, Saswat Anand, and Alex Aiken. Minimizing GUI event traces. *FSE*, 2016.

Osbert Bastani, Saswat Anand, and Alex Aiken. Interactively verifying absence of explicit information flows in android apps. *OOPSLA*, 2015.

Osbert Bastani, Saswat Anand, and Alex Aiken. Specification inference using context-free language reachability. *POPL*, 2015.

Osbert Bastani, Christopher J Hillar, Dimitar Popov, and J Maurice Rojas. Randomization, sums of squares, near-circuits, and faster real root counting. *Contemporary Mathematics*, 556:145–166, 2011.

Invited Talks

2026 **MIT Media Lab Advancing Humans with AI**.

2026 **Toyota Technical Institute at Chicago Seminar**.

2026 **SIAM Conference on Optimization**.

2026 **University of Chicago CS Seminar**.

2025 **Simons Workshop on Theoretical Aspects of Trustworthy AI**.
2024 **Worskhop on Interpretable Policies in Reinforcement Learning at RLC**.
2024 **Cornell AI Seminar**.
2024 **Cornell Tech NLP Seminar**.
2024 **Stanford MLSys Seminar**.
2024 **Pennsylvania State University Systems, Controls, and Robotics Seminar**.
2023 **University of British Columbia TrustML Workshop**.
2023 **ACC Safe Perception-based Control Workshop**.
2023 **University of Pennsylvania ASSET Seminar**.
2023 **IPAM Workshop on Explainable AI for the Sciences: Towards Novel Insights**.
2022 **ICML Workshop on Distribution-Free Uncertainty Quantification**.
2022 **University of Pennsylvania CIS Colloquium**.
2022 **Yale CS Seminar**.
2021 **IFDS Summer Workshop on Statistical Approaches to Understanding Modern Machine Learning Methods**.
2021 **Simons Institute Workshop on Games and Equilibria**.
2020 **ICML Workshop on Explainable AI: Beyond Deep Models and Classifiers**.
2020 **UCSD AI Seminar**.
2020 **UCSB Programming Languages Seminar**.
2019 **UT Austin Programming Languages Seminar**.
2019 **ICLR Workshop on Debugging Machine Learning Models**.
2018 **CPS Week Workshop on Design and Analysis of Robust Systems**.
2017 **University of Pennsylvania Computer Science Seminar**.
2017 **Cornell University Computer Science Colloquium**.
2017 **Northwestern University Computer Science Seminar**.
2017 **Penn State University Computer Science Seminar**.

Funded Grants

2026–2027 **PI**, *Measuring Human Agency on Digital Platforms for Social Impact*.
2026–2030 **Co-PI**, *RI: Uncertainty Quantification for Robust Decision Making*.
2024–2028 **Co-PI**, *SCH: Robust and Equitable Clinical Decision Support in Glaucoma Detection and Progression Prediction*.
2024–2025 **PI**, *Amazon Research Award: Uncertainty Quantification for Trustworthy Language Generation*.
2024–2029 **PI**, *NSF CAREER: Formal Guarantees for Neurosymbolic Programs via Conformal Prediction*.
2023–2027 **Co-PI**, *NSF SLES: SPECSRL: Specification-guided Perception-enabled Conformal Safe Reinforcement Learning*.
2020–2025 **Co-PI**, *NSF Expeditions: Understanding the World Through Code*.
2020–2025 **Co-PI**, *ARO MURI: Robust Concept Learning and Lifelong Adaptation Against Adversarial Attacks*.
2019–2023 **PI**, *NSF SHF: Small: Inferring Specifications for Blackbox Code*.
2019–2022 **Co-PI**, *DARPA Learning with Less Labels*.

Ph.D. Student Advising

2026– **Frank Li**, *Current student*.
2024– **Marcus Min**, *Current student*.

2024– **Sagnik Anupam (co-advised with Dinesh Jayaraman)**, *Current student*.

2024– **Emma Shedden (co-advised with Rajeev Alur)**, *Current student*.

2022– **Alex Shypula**, *Current student*.

2022– **Yimeng Zeng (co-advised with Jake Gardner)**, *Current student*.

2021– **Ramya Ramalingam (co-advised with Aaron Roth)**, *Current student*.

2019–2026 **Stephen Mell (co-advised with Steve Zdancewic)**, *Thesis: Opportunistic Evaluation for Compound AI Systems*.

2023–2025 **Michael Yao (co-advised with James Gee)**, *Thesis: Distributionally Robust Machine Intelligence for Medicine and Scientific Discovery*.

2020–2025 **Jason Ma (co-advised with Dinesh Jayaraman)**, *Thesis: Foundation Reward Models for Robot Learning*.

2020–2025 **Shuo Li (co-advised with Insup Lee)**, *Thesis: Towards Trustworthy Large Language Models*.

2019–2024 **Halley Young**, *Thesis: Specification Guided Generative Models*.

2019–2021 **Sangdon Park (co-advised with Insup Lee)**, *Thesis: Uncertainty Estimation Toward Safe AI*.

Undergraduate/Masters Student Advising

2026– **Christina Cao (undergraduate)**.

2026– **Nithya Srikumar (masters)**.

2026– **Garrett Lee (masters)**.

2026– **Jensen Liu (masters)**.

2026– **Yifei Li (masters)**.

2026– **Tony Quan (masters)**.

2026– **Ssaumya Jaiswal (masters)**.

2026– **Zheng Xing (undergraduate)**.

2026– **Calvin Zhou (undergraduate)**.

2025–2026 **One An (undergraduate)**.

2025–2026 **Aneesh Edara (undergraduate)**.

2025–2026 **Utkarsh Lal (masters)**.

2024–2026 **Remy Kim (masters)**, *Master's Thesis: Belief Representations for Sequential Decision Making under Uncertainty in Neurosymbolic Systems*.

2024–2026 **Lianghuan Huang (undergraduate)**.

2023–2026 **Botong Zhang (undergraduate)**.

2024–2026 **Nathan Yu (undergraduate)**.

2024–2025 **Tara Kapoor (undergraduate)**.

2022–2023 **Adam Khakhar (undergraduate)**.

2021–2024 **Estelle Shen (undergraduate)**.

2020–2024 **Sarah Luthra (undergraduate)**.

2023–2024 **Divya Kumari (masters)**.

2023 **Ranbir Mahtani (undergraduate)**, *Senior Thesis: GPT's Theory, Performance, Applications, and Future Direction*.

2021–2022 **Angelina Heyler (masters)**, *Master's Thesis: PAC Prediction Sets for Deep Neural Networks Trained via Federated Learning*.

2021–2022 **Aishwarya Wesanekar (masters)**.

2021–2022 **Utkarsh Kashyap (masters)**.

2020–2022 **Ryan Gannon (undergraduate)**.
2020–2022 **Aryan Singh (undergraduate)**.
2020–2021 **Jian Zhang (undergraduate)**.
2020–2021 **George Tolkachev (masters)**.
2020–2021 **Wanqiao Xu (undergraduate)**.
2020 **Maxwell Du (undergraduate)**.
2019 **Brian Heath (masters)**.

Service (Penn)

2025– **Penn Master of Science and Engineering (MSE) in Data Science Program**, Advisor.
2025–2026 **Penn Pilot Fellowship Review Panel**, Reviewer.
2025–2026 **Penn Department of Computer and Information Science Ph.D. Admissions**, Coordinator.
2024 **Penn Departmental of Computer and Information Science Colloquium**, Organizer.

Service (External)

2027 **ICLR 2027**, Senior Area Chair.
2026 **NeurIPS 2026**, Area Chair.
2026 **COLM 2026**, Area Chair.
2026 **NeuS 2026**, Program Committee.
2026 **ICML 2026**, Area Chair.
2026 **OOPSLA 2026**, Program Committee Member.
2025 **MLSys 2026**, Program Committee Member.
2025 **ICLR 2026**, Area Chair.
2025 **ASPLOS 2026**, Program Committee Member.
2024 **ICLR 2025**, Area Chair.
2024 **ASPLOS 2025**, Program Committee Member.
2023 **ICLR 2024**, Area Chair.
2023 **AAAI 2024**, Senior Program Committee Member.
2023 **OOPSLA 2023**, Review Committee Member.
2022 **AAAI 2023**, Senior Program Committee Member.
2022 **OOPSLA 2022**, Program Committee Member.
2021 **PLDI 2022**, Program Committee Member.
2020 **PLDI 2021**, Program Committee Member.
2019 **POPL 2020**, Program Committee Member.
2019 **CAV 2019**, Program Committee Member.
2018 **PLDI 2019**, External Program Committee Member.
2017 **PLDI 2018**, External Program Committee Member.

Teaching

Spring 2026 **CIS 4270/5270**, *University of Pennsylvania*, Trustworthy Machine Learning.
Spring 2025 **CIS 4190/5190**, *University of Pennsylvania*, Applied Machine Learning.
Spring 2024 **CIS 7000**, *University of Pennsylvania*, Trustworthy Machine Learning.
Fall 2023 **CIS 4190/5190**, *University of Pennsylvania*, Applied Machine Learning.
Spring 2023 **CIS 4190/5190**, *University of Pennsylvania*, Applied Machine Learning.